Dual-Role Controls in Dynamic Panels: Quantifying the Cost of Including or Excluding Time-Varying Covariates

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Abstract

Researchers using panel data routinely face a covariate inclusion dilemma: the same time-varying variable Z can be both a confounder (omitting it biases the estimate) and a collider or mediator (including it biases the estimate in a different direction). We introduce the Included Variable Bias (IVB) formula, a diagnostic that quantifies the change in the treatment estimate from including or excluding a covariate, using quantities the researcher already has. The formula mirrors the classic Omitted Variable Bias formula but addresses the opposite problem. For modern panel estimators with adaptive weights, such as Synthetic Difference-in-Differences (SDiD), we decompose IVB into a Coefficient Effect (CE) and a Reweighting Effect (RE), where CE preserves the IVB product structure and RE captures weight distortion from covariate inclusion. Applied to [NUMBER] published studies, the diagnostic transforms an opaque specification choice into a quantified, transparent trade-off.

1 Introduction

Should a time-varying covariate be included in a panel regression? Consider a scholar estimating the effect of UN peacekeeping on democratization. Conflict intensity predicts both the deployment of peacekeeping missions and the prospects for democratic transition, so omitting it risks confounding bias. But peacekeeping can itself reduce violence, making realized conflict a consequence of treatment. Including it opens a different source of bias. The researcher must choose: omit the covariate and risk omitted variable bias, or include it and risk what we call included variable bias.

This is not a special case. GDP, trade flows, foreign aid, inflation, and dozens of other variables evolve

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alongside treatment and outcome in panel data. Researchers routinely decide whether to condition on them, and the decision can move the point estimate. The Omitted Variable Bias formula gives researchers a way to assess the cost of excluding a confounder. No analogous tool exists for the cost of including a covariate that may be post-treatment, a collider, or a mediator. We provide one.

We introduce the Included Variable Bias (IVB) formula:

$$\text{IVB} = -\theta^{\star} \times \pi \tag{1}$$

where $\theta^{\star}$ is the coefficient on the covariate in the augmented regression and π is the coefficient on the treatment in an auxiliary regression of the covariate on the treatment and legitimate controls. Both quantities come from regressions the researcher has already run. Where OVB measures the cost of leaving a variable out, IVB measures the change from putting one in. Combined with a causal graph, the formula tells the researcher whether the change represents collider bias, confounding removal, or over-control of a mediator, and how large it is. We derive the formula for two-way fixed effects, autoregressive distributed lag, and difference-in-differences specifications with time-varying covariates.

A growing literature on difference-in-differences, synthetic control, and synthetic difference-in-differences recognizes that parallel trends often hold only conditionally on covariates (Sant'Anna and Zhao 2020; Callaway and Sant'Anna 2021; Roth et al. 2023), and recent work develops methods for incorporating covariates optimally into these estimators (Botosaru and Ferman 2019; Hirshberg and Klosin 2024; Arkhangelsky and Imbens 2024). These contributions assume the covariates are exogenous. When a covariate needed for identification may also respond to treatment, the dual-role dilemma reappears in a setting where no diagnostic exists. Estimators that construct counterfactuals using adaptive weights compound the problem: including a covariate changes which units and periods form the comparison group, not just the regression coefficients. We extend IVB to this class of estimators by decomposing the total sensitivity into a Coefficient Effect, which preserves the product structure of the formula above, and a Reweighting Effect, which captures weight distortion from covariate inclusion.

We apply the diagnostic to [NUMBER] published studies in [FIELDS], identify covariates with plausible dual roles, and assess whether the specification choice is consequential. The applications show when IVB is small enough to ignore and when the decomposition reveals offsetting forces invisible from comparing point estimates alone.

Section~2 derives the IVB formula and discusses dual-role covariates in dynamic settings. Section~3 extends the decomposition to estimators with adaptive weights. Section~4 applies the diagnostic to published studies.

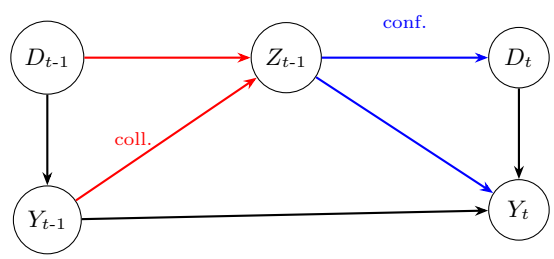
Section~5 concludes.

2 The IVB Formula

2.1 Dual-Role Covariates in Dynamic Settings

The covariate inclusion dilemma described in the introduction has a structural basis in dynamic settings. In cross-sectional work with directed acyclic graphs (DAGs), a variable cannot simultaneously be a confounder and a collider: the first requires $Z \to D$ or $Z \to Y$; the second requires $D \to Z \leftarrow Y$. Both together imply a cycle, which acyclic graphs exclude by construction. A variable is one or the other; the researcher's task is classification, not quantification of a trade-off.

In TSCS data, the temporal dimension dissolves this constraint. The same variable Z can be a confounder in one period and a collider in another, without creating cycles in the time-indexed DAG:



Here Z_{t-1} is a confounder for the effect $D_t \to Y_t$ (blue arrows: $Z_{t-1} \to D_t$ and $Z_{t-1} \to Y_t$) and simultaneously a collider opened by D_{t-1} and Y_{t-1} (red arrows). No cycle exists because the nodes are indexed by time. Conditioning on Z_{t-1} removes confounding but opens a collider path. The researcher faces a genuine trade-off, and the magnitude of that trade-off depends on the data. We note that dual-role variables can also arise outside the acyclic framework, for instance in simultaneous equations or network settings; our argument is specific to the acyclic graphs that dominate applied practice in political science.

A practical difficulty compounds the problem. Establishing that a covariate is a collider often requires consulting literatures outside the researcher's own field. We call this *foreign collider bias*: the risk of conditioning on a collider whose status depends on causal evidence from adjacent disciplines. A political scientist studying whether democracy reduces ethnic inequality may include GDP per capita as a control, since wealth correlates with both democracy and equality. The collider channels are visible only from adjacent fields: growth economists estimate that democratization raises GDP by roughly 20% over 25 years (Acemoglu et al. 2019); development economists find that ethnic inequality reduces GDP per capita by 12–15% (Grundler and Link 2024). Both $D \to Z$ and $Y \to Z$ are active, making GDP a collider. The

researcher who does not consult both literatures will not realize it.

The IVB formula provides a way to assess the consequences of this uncertainty. Even when the researcher cannot fully resolve the DAG, the formula quantifies how much the estimate changes from including or excluding the covariate, and decomposes that change into interpretable components.

2.2 The Formula

To build intuition, consider the cross-sectional case. In a regression of y on D (`short'' model), adding a variable $Z$ produces the`long'' model with coefficients $\beta_1^\star$ (on D) and $\beta_2^\star$ (on Z). The change in the treatment coefficient satisfies IVB $= \beta_1^\star - \beta_1 = -\beta_2^\star \times \phi_1$, where ϕ_1 is the coefficient from regressing Z on D. This is the standard short-versus-long regression decomposition, well known in econometrics [e.g., Goldberger (1991), pp. 197–198; see Appendix~5.1 for details]. Table~1 presents the comparison with OVB.

Table 1: Comparison of Omitted Variable Bias (OVB) and Included Variable Bias (IVB)

	OVB	**IVB**
Problem	Omitting a confounder	Including a collider
Correct model	Long (includes confounder)	Short (excludes collider)
Wrong model	Short (omits confounder)	Long (includes collider)
Formula	$\delta_1 \times \gamma$	$-\beta_2^\star \times \phi_1$
Components	Effect of omitted var in long reg	Effect of included var in long reg
	Regression of omitted var on D	Regression of included var on D
Sign of bias	$\text{sign}(\delta_1) \times \text{sign}(\gamma)$	$-\text{sign}(\beta_2^\star) \times \text{sign}(\phi_1)$
Estimable?	Only if confounder is observed	Yes, both components from data

A notable practical advantage of IVB is that its components are *always* estimable: the researcher has already included Z in the regression, so $\beta_2^\star$ is available, and ϕ_1 comes from a simple auxiliary regression. By contrast, OVB requires observing the omitted variable—which, by definition, is often unavailable.

2.3 Panel Fixed Effects

The cross-sectional identity extends to panel models, where it becomes a new diagnostic tool. Consider the short model:

$$y_{it} = \alpha_i + \mu_t + \beta D_{it} + e_{it} \tag{2}$$

and the augmented model including a time-varying covariate Z_{it}:

$$y_{it} = \alpha_i^\star + \mu_t^\star + \beta^\star D_{it} + \theta^\star Z_{it} + v_{it} \tag{3}$$

By the Frisch-Waugh-Lovell theorem, unit and time fixed effects are absorbed by projecting all variables onto the space spanned by the unit and time dummies. The cross-sectional decomposition applies to the doubly-demeaned residuals:

Proposition 1 (IVB under TWFE)**.** *In a two-way fixed effects model, the change in the treatment effect from including Z_{it} is:*

$$IVB = \beta^\star - \beta = -\theta^\star \times \pi \tag{4}$$

where $\theta^\star$ is the coefficient on Z_{it} in the long TWFE regression and π is the coefficient on D_{it} in the auxiliary TWFE regression $Z_{it} = \alpha_i' + \mu_t' + \pi D_{it} + \eta_{it}$.

The corollary has a substantive implication. Because unit fixed effects absorb any time-invariant component of the association between D and Z, the auxiliary coefficient π reflects only the *within-unit* co-movement of treatment and covariate. If D and Z are correlated primarily in levels across units, unit fixed effects remove this correlation and π is small. This is one reason why IVB tends to be modest in panel applications: fixed effects absorb between-unit collider channels.

2.4 Extension to Dynamic Models

The formula extends to any linear dynamic specification via FWL. In an ADL(1,0) model with the legitimate controls $W_t = (1, y_{t-1})$, residualizing on W_t and applying the same decomposition yields:

Remark 1 (IVB in dynamic models)**.** *For any linear specification including ADL(p,q) with unit and time fixed effects, the IVB for any treatment coefficient β_j takes the form $IVB(\beta_j) = -\theta^\star \times \pi_j$, where π_j is the coefficient on D_{t-j} in the auxiliary regression of Z on the treatment and all legitimate controls. The form is uniform across specifications; see Appendix 5.2 for the full derivation.*

The practical value lies in this uniformity. Regardless of the number of lags in the outcome or the treatment, the IVB for any coefficient is always the negative product of two estimable quantities: the covariate coefficient $\theta^\star$ in the long model and the partial association π_j between the treatment and the covariate, controlling for legitimate regressors.

2.5 A Practical Recipe

We summarize the IVB formula as a diagnostic procedure:

Step 0. Assess causal status. Using a DAG or substantive expertise, assess whether Z is plausibly a confounder, a collider, a mediator, or a dual-role variable. The formula is informative regardless—it quantifies the specification shift—but the DAG determines the *interpretation*.

1. **Run the long regression.** Estimate the model that includes Z; record $\hat{\theta}^{\star}$.
2. **Run the auxiliary regression.** Regress Z on the treatment and all legitimate controls; record $\hat{\pi}$.
3. **Compute the IVB.** $\widehat{\text{IVB}} = -\hat{\theta}^{\star} \times \hat{\pi}$. Compare to $\text{SE}(\hat{\beta})$ to assess materiality. See Appendix 5.3 for uncertainty quantification.

2.6 Interpretation

The decomposition $\text{IVB} = -\theta^{\star} \times \pi$ is an algebraic identity. It holds whenever the short and long regressions are nested, regardless of the causal role of Z. The three cases lead to different conclusions:

- Z **is a collider.** The short model is correct; IVB is the bias from including Z. Exclude Z.
- Z **is a confounder.** The long model is correct; the same decomposition gives the OVB in the short model. Include Z.
- Z **is a dual-role variable.** Both channels are active. The IVB formula quantifies the *net* specification shift, but the causal interpretation requires additional reasoning. In TSCS settings, conditioning on the lagged value Z_{t-1} often resolves the tension: it captures confounding without opening the contemporaneous collider channel.

The formula does not substitute for causal reasoning. It complements DAGs by quantifying what classification alone cannot: the *magnitude* of the specification choice.

3 IVB in Modern Panel Estimators

3.1 Motivation

The IVB formula from Section~2 applies to any linear specification: TWFE, ADL, DiD with covariates. In all these cases, the "weights'' on observations are determined by the OLS projection and do not change when covariates are added or removed. The IVB is entirely a coefficient-channel phenomenon.

Modern panel estimators—Synthetic Difference-in-Differences [SDiD; Arkhangelsky et al. (2021)], Synthetic Control, Interactive Fixed Effects—use *adaptive* weights that are jointly optimized with the treatment effect and covariate coefficients. Including a covariate Z changes not only the regression coefficients but also *which units and periods* form the counterfactual. This creates a new source of specification sensitivity with no TWFE analogue.

We extend the IVB framework to SDiD by decomposing the total sensitivity into two components: a Coefficient Effect (CE) that preserves the familiar product structure, and a Reweighting Effect (RE) that captures weight distortion.

3.2 Setup and Notation

Consider a balanced panel with N units (N_0 controls, N_1 treated) observed over T periods (T_0 pre-treatment, T_1 post-treatment). The SDiD estimator uses unit weights $\omega \in \Delta_{N_0}$ and time weights $\lambda \in \Delta_{T_0}$, optimized to balance pre-treatment trends. Define the extended weight vectors:

$$\tilde{\omega} = (-\omega_1, \ldots, -\omega_{N_0}, 1/N_1, \ldots, 1/N_1)' \in \mathbb{R}^N \tag{5}$$

$$\tilde{\lambda} = (-\lambda_1, \ldots, -\lambda_{T_0}, 1/T_1, \ldots, 1/T_1)' \in \mathbb{R}^T \tag{6}$$

The SDiD estimate without covariates is the weighted double-difference $\hat{\tau}_{\text{short}} = \tilde{\omega}'_{\text{short}} \mathbf{Y} \tilde{\lambda}_{\text{short}}$. With a covariate Z, the estimator jointly optimizes weights and the covariate coefficient: $\hat{\tau}_{\text{long}} = \tilde{\omega}'_{\text{long}}(\mathbf{Y} - \mathbf{Z}\hat{\beta}_{\text{long}})\tilde{\lambda}_{\text{long}}$.

3.3 The Decomposition

Definition 1 (Three estimators). 1. ***Short***: *$\hat{\tau}_{short}$, SDiD without Z, optimized weights $(\omega_{short}, \lambda_{short})$.*

2. ***Long***: *$\hat{\tau}_{long}$, SDiD with Z, jointly optimized $(\omega_{long}, \lambda_{long}, \hat{\beta}_{long})$.*

3. ***Hybrid***: *$\hat{\tau}_{hybrid}$, SDiD with Z but weights* fixed *at $(\omega_{short}, \lambda_{short})$. Only $\hat{\beta}_h$ is optimized.*

The hybrid estimator is the bridge. It isolates the effect of adding Z to the regression from the effect of Z on weight optimization:

$$\boxed{\text{IVB} = \underbrace{(\hat{\tau}_{\text{hybrid}} - \hat{\tau}_{\text{short}})}_{\text{CE: Coefficient Effect}} + \underbrace{(\hat{\tau}_{\text{long}} - \hat{\tau}_{\text{hybrid}})}_{\text{RE: Reweighting Effect}}} \tag{7}$$

The decomposition is exact by construction.

Proposition 2 (FWL-like identity for CE)**.** *Under SDiD with a scalar covariate Z and fixed weights:*

$$CE = -\hat{\beta}_h \times \hat{\tau}_Z \tag{8}$$

where $\hat{\beta}_h$ is the coefficient of Z in the hybrid model and $\hat{\tau}_Z = \tilde{\omega}'_{short}\mathbf{Z}\tilde{\lambda}_{short}$ is the SDiD double-difference applied to Z.

Proof. Both $\hat{\tau}_{\text{hybrid}}$ and $\hat{\tau}_{\text{short}}$ use the same weights. By bilinearity:

$$\text{CE} = \tilde{\omega}'(\mathbf{Y} - \mathbf{Z}\hat{\beta}_h)\tilde{\lambda} - \tilde{\omega}'\mathbf{Y}\tilde{\lambda} = -\hat{\beta}_h \cdot \tilde{\omega}'\mathbf{Z}\tilde{\lambda} = -\hat{\beta}_h \times \hat{\tau}_Z \tag{9}$$

□

The identity maps directly to the TWFE formula: $\hat{\beta}_h$ is the SDiD analogue of $\theta^\star$, and $\hat{\tau}_Z$ is the analogue of π. With multiple covariates, the identity generalizes to $\text{CE} = -\sum_k \hat{\beta}_{h,k} \times \hat{\tau}_{Z_k}$.

Corollary 1 (TWFE as special case)**.** *When weights are fixed (not adaptive), $RE = 0$ and $IVB = CE$. TWFE is the limiting case where the Reweighting Effect vanishes.*

3.4 When Is RE = 0?

Proposition 3 (Additive Z implies RE = 0)**.** *Assume SDiD with intercepts. If $Z_{it} = a_i + b_t$ for some unit-specific a_i and time-specific b_t, then $\omega_{long} = \omega_{short}$, $\lambda_{long} = \lambda_{short}$, and $RE = 0$.*

The proof (Appendix~5.4) uses two properties: the simplex constraint ($\sum \omega_i = \sum \lambda_t = 1$) causes common components to cancel, and the intercepts absorb remaining level shifts. The result implies that covariates capturing permanent unit characteristics ($Z_{it} = a_i$) or common shocks ($Z_{it} = b_t$) do not distort the synthetic weights. Only the *interactive* component of Z—the within-unit-and-time residual after removing unit and time effects—can induce RE.

A practical pre-diagnostic follows: regress Z on unit and time fixed effects. If R^2 is high, RE is expected to be small and IVB $\approx$ CE.

3.5 Diagnostic Protocol for SDiD

Step 1. Pre-diagnostic. Regress $Z \sim \alpha_i + \mu_t$. If $R^2_Z > 0.90$, expect RE ≈ 0.

Step 2. Compute three estimates. Run `synthdid_estimate` without Z (short), with Z (long), and with Z but weights fixed from the short model (hybrid).

Step 3. Decompose. $\text{CE} = \hat{\tau}_{\text{hybrid}} - \hat{\tau}_{\text{short}}$, $\text{RE} = \hat{\tau}_{\text{long}} - \hat{\tau}_{\text{hybrid}}$. Verify $\text{CE} = -\hat{\beta}_h \times \hat{\tau}_Z$.

Step 4. Report.

	Value	Share of IVB
$\hat{\tau}_{\text{short}}$ (without Z)		
$\hat{\tau}_{\text{long}}$ (with Z)		
IVB		100%
CE (coefficient effect)		%
RE (reweighting effect)		%
$\hat{\beta}_h$ ($Z \to Y$)		
$\hat{\tau}_Z$ ($D \to Z$)		

Interpretation. If |IVB| is small relative to SE($\hat{\tau}$): specification choice is immaterial. If |IVB| is large and CE-dominated: the change is through the coefficient channel—reason about whether $Z \to Y$ and $D \to Z$ reflect confounding or mediation. If RE-dominated: weight distortion is the main driver—examine whether the reweighted counterfactual has better or worse pre-treatment fit.

4 Applications

[PLACEHOLDER: This section will apply the IVB diagnostic to published studies, following the approach of @oster2019unobservable. Studies will be selected based on: (1) use of TWFE, DiD, or SDiD with time-varying covariates; (2) covariates with plausible dual-role status; (3) replication data availability. The protocol from Sections 2.5 and 3 will be applied to each study. For SDiD applications, the CE + RE decomposition will be reported. The selection of specific studies is pending a systematic search.]

5 Conclusion

Researchers using panel data routinely include time-varying covariates without a framework to assess the consequences. The same variable can be a confounder, a collider, or a mediator—and in dynamic settings, can play multiple roles across periods without violating the acyclicity of the causal graph. We provide a diagnostic that quantifies the cost of the specification choice.

The IVB formula, $\text{IVB} = -\theta^\star \times \pi$, expresses the change in the treatment estimate from including or excluding a covariate as the negative product of two quantities the researcher already has. It applies to TWFE, ADL, and DiD models. For SDiD estimators, the decomposition $\text{IVB} = \text{CE} + \text{RE}$ further separates the coefficient channel from the weight distortion channel, revealing structure that is invisible in TWFE.

The formula does not resolve the causal graph—it does not tell the researcher whether a covariate is a confounder or a collider. What it does is transform an opaque specification choice into a quantified, transparent trade-off. The researcher sees how much the estimate changes, through which channel, and which covariates drive the change. This is valuable whether or not the DAG is fully resolved.

Several limitations should be noted. The formula applies to the contemporaneous treatment effect; cumulative and dynamic effects require different tools (e.g., Blackwell and Glynn 2018). The derivation assumes linear models; extensions to nonlinear estimators are left to future work. The SDiD decomposition requires the `synthdid` package's support for fixed weights, which may not be available in all implementations. Finally, the diagnostic is a sensitivity analysis, not a specification test—it quantifies but does not prescribe.

Future work could develop a sensitivity framework in the spirit of Oster (2019) and Cinelli and Hazlett (2020), asking: "how large would $\theta^\star$ or π need to be to overturn the result?" Extensions to Interactive Fixed Effects and Generalized Synthetic Control would complete the coverage of modern panel estimators.

Appendix

5.1 A. Cross-Section Derivation

[Full cross-section derivation of IVB = $-\beta_2^\star \times \phi_1$. Reuse from existing paper Appendix A.]

5.2 B. ADL Derivation

[Full FWL-based derivation for ADL(1,0) and ADL(p,q). Reuse from existing paper Appendix B.]

5.3 C. Variance of IVB Estimator

[Delta method variance and conservative upper bound. Reuse from existing paper Appendix G.]

5.4 D. SDiD Derivation Details

[Full proof of Proposition 3. Collapsed form preserves bilinear structure. Adapted from derivation note.]

5.5 E. Nickell Bias Interaction

[IVB formula is exact as identity even with Nickell bias. Reuse from existing paper Appendix C.]

5.6 F. Lag Substitution

[Replacing D_t with D_{t-k} does not mechanically eliminate IVB. Proposition 4 and Corollary 2. Reuse from existing paper Appendix D.]

5.7 G. IVB in Potential Outcomes / DiD Decomposition

[Re-derivation in PO notation. Connection to Caetano & Callaway. DiD decomposition. Reuse from existing paper Appendices F and Section 6.]

5.8 H. Full Application Results

[Control classification tables and full IVB results for all studies. Reuse from existing paper Appendix I.]

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